

ELECTROMAGNETIC FORM FACTORS OF THE PROTON AT LOW FOUR-MOMENTUM TRANSFER

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Abstract: Electron-proton elastic scattering cross sections were measured at low four-momentum transfers squared (q^2 from 0.13 to 2.15 fm⁻²) at six different energies between 150 and 275 MeV. The electric (G_E) and magnetic (G_M) form factors of the proton have been determined by Rosenbluth plots and independently by using analytical functions for the form factors to fit the cross sections. The electric form factor is found to deviate significantly from the dipole fit. From the slope of the form factor functions at $q^2 = 0$ the rms radii of the charge and the magnetic moment distribution were determined. The charge rms radius is found to be more than 10% larger than the value given by the dipole fit.

E

NUCLEAR REACTIONS p(e, e), $E = 150\text{--}275$ MeV; measured $\sigma(E_e, \theta)$. Proton deduced electromagnetic form factors. Liquid target.

1. Introduction

The description of electron-proton elastic scattering cross sections in terms of the electric and magnetic structure form factors $G_E(q^2)$ and $G_M(q^2)$, based on the approximation of a single photon exchange process, is given by the Rosenbluth formula ¹⁾. The sensitivity of the elastic e-p scattering cross sections to the electric form factor G_E increases rapidly with decreasing q^2 . In the kinematic region of the present experiment the electric form factor strongly dominates the magnetic part in the Rosenbluth formula, thus allowing a precise determination of $G_E(q^2)$.

The rms radii of the spatial distributions of charge and magnetic moment of the proton are interesting because they occur often in nuclear and high energy physics. The rms radii are given by the q^2 dependence of the form factors at $q^2 = 0$:

$$\langle r_{E,M}^2 \rangle = - \frac{6}{G_{E,M}(0)} \frac{dG_{E,M}(q^2)}{dq^2} \bigg|_{q^2=0}. \quad (1)$$

The determination of the form factors in a relatively wide q^2 range near $q^2 = 0$ allows a meaningful extrapolation to the photon point which is necessary to determine the radii.

The q^2 dependence of the form factors is reasonably well approximated in a large q^2 range by the empirical dipole fit and the form factor scaling relation ²⁾

$$\begin{aligned} G_D(q^2) &= 1/(1+q^2/18.23)^2, \\ G_E(q^2) &= G_M(q^2)/\mu, \end{aligned} \quad (2)$$

where q^2 is in fm^{-2} and μ is the magnetic moment of the proton. The dipole fit gives $\langle r_E^2 \rangle^{\frac{1}{2}} = \langle r_M^2 \rangle^{\frac{1}{2}} = 0.81 \text{ fm}$.

Previous measurements at Bonn³⁾ and DESY⁴⁾ show distinct deviations from the dipole fit at high q^2 and indicate deviations in the low q^2 region. These deviations have been confirmed by the present low q^2 experiment. From these experiments an improved fit to the form factors has been obtained. This fit shows that $G_E(q^2)$ at low q^2 decreases faster than the dipole fit which may be meaningful for theoretical calculations. Furthermore the new fit may be used to calculate reference cross sections for medium energy electron scattering experiments on nuclei.

2. Measurements

Proton elastic scattering cross sections have been measured using the experimental facilities of the Mainz 300 MeV electron linear accelerator. As the arrangement has been described in detail previously⁵⁾ we only give the essential features of the present experiment.

The cross sections were taken at six different energies between 150 MeV and 275 MeV at angles from 28° to 75° corresponding to a range of the momentum transfer squared q^2 from 0.13 to 2.15 fm^{-2} . A cryogenic target system was installed in the scattering chamber. It contained a cylindrical liquid hydrogen cell (10 mm in diameter, 12 μm capton walls), a similar empty target cell, a carbon target and a fluorescent viewer, which alternately could be placed at beam position. The carbon target was used for a precise determination of the energy of the incident electrons ($\pm 0.2\%$) via the sharp peak of the elastically scattered electrons. The momentum spread of the incoming electron beam was 0.1%.

Most of the experimental points were taken with a 180° double focusing spectrometer, set to different scattering angles. The scattered electrons were recorded by a 75 channel ladder counter system with a total momentum acceptance of 7.5%. Thus the shape of the radiative tail was detected over a wide range with high resolution which was useful for a check of the radiative corrections.

Additionally at each energy an absolute cross section was measured with a second magnetic spectrometer installed at a fixed angle of 28° [ref. ⁶⁾]. The solid angle of this spectrometer was defined by the entrance slit. The efficiency of the detection system was higher than 99.5%. The target thickness was determined from the diameter of the target cell and the temperature of the liquid hydrogen. As the scattering angle was small the measurements could be performed at low beam intensities which is essential for a reliable determination of the target thickness.

The cross sections measured with the first spectrometer were taken in different runs at a constant electron energy. Simultaneously the counting rate of the second spectrometer was recorded. These results at a fixed energy and a fixed angle were used as an indication of possible fluctuations of the target density during each run. Thus the data of the first spectrometer are independent from changes in the target thickness if the counting rate is normalized to the reading of the second spectrometer.

TABLE 1
The electron-proton elastic cross sections obtained from this experiment

Incident beam energy (MeV)	Scattering angle θ (deg)	q^2 (fm $^{-2}$)	Cross section $d\sigma/d\Omega$ (10^{-30} cm 2 /sr)	Random errors ($\pm$)	$d\sigma/d\Omega / \frac{d\sigma}{d\Omega_D}$	Errors of the ratio ($\pm$)
149.8	28.0	0.13	60.9	1.2	0.994	0.020
	40.0	0.26	13.81	0.13	0.992	0.009
	45.0	0.32	8.22	0.08	0.974	0.009
	50.0	0.39	5.27	0.06	0.977	0.011
	55.0	0.46	3.49	0.03	0.977	0.009
	60.0	0.53	2.40	0.02	0.980	0.007
	65.0	0.61	1.668	0.014	0.987	0.009
	70.0	0.69	1.197	0.012	0.965	0.010
	75.0	0.76	0.871	0.010	0.956	0.010
180.1	28.0	0.19	41.8	0.8	0.997	0.019
	40.0	0.37	9.28	0.08	0.981	0.008
	45.0	0.46	5.55	0.05	0.970	0.008
	50.0	0.56	3.56	0.03	0.980	0.008
	55.0	0.66	2.316	0.017	0.964	0.007
	60.0	0.76	1.588	0.011	0.969	0.006
	65.0	0.86	1.107	0.008	0.963	0.007
	70.0	0.97	0.799	0.007	0.966	0.010
	75.0	1.08	0.586	0.006	0.965	0.010
199.6	28.0	0.23	33.2	0.7	0.982	0.019
	40.0	0.46	7.32	0.10	0.967	0.012
	45.0	0.56	4.44	0.05	0.972	0.011
	50.0	0.68	2.78	0.04	0.959	0.013
	60.0	0.93	1.247	0.014	0.960	0.011
	70.0	1.18	0.624	0.007	0.955	0.011
228.9	28.0	0.31	24.8	0.5	0.975	0.019
	40.0	0.60	5.51	0.07	0.981	0.013
	45.0	0.74	3.26	0.04	0.965	0.012
	50.0	0.88	2.06	0.02	0.967	0.011
	55.0	1.04	1.322	0.016	0.945	0.011
	60.0	1.19	0.905	0.009	0.953	0.009
	70.0	1.53	0.458	0.004	0.962	0.009
	75.0	1.69	0.337	0.003	0.964	0.008
249.6	28.0	0.36	20.2	0.4	0.957	0.019
	35.0	0.55	7.89	0.09	0.958	0.011
	40.0	0.71	4.46	0.06	0.961	0.012
	45.0	0.87	2.66	0.03	0.956	0.010
	50.0	1.04	1.653	0.023	0.945	0.013
	55.0	1.23	1.079	0.015	0.942	0.014
	60.0	1.41	0.734	0.008	0.946	0.011
	65.0	1.60	0.517	0.007	0.956	0.013
	70.0	1.79	0.366	0.004	0.944	0.011
275.3	75.0	1.98	0.271	0.004	0.953	0.012
	28.0	0.44	16.84	0.34	0.982	0.020
	50.0	1.26	1.328	0.011	0.954	0.009
	60.0	1.70	0.578	0.006	0.943	0.011
	70.0	2.15	0.290	0.003	0.950	0.010

The dipole cross sections $d\sigma/d\Omega_D$ were obtained using the Rosenbluth formula and eq. (2).

The data were corrected for empty target background and finite vertical and horizontal angular acceptance of the spectrometer. In calculating the cross sections the energy spectra of the scattered electrons were integrated down to $E_{\text{elastic}} - \Delta E$, where ΔE is the fivefold half-width of the peak. The main corrections which have been applied are the real bremsstrahlung correction due to Mo and Tsai ⁷⁾ and the radiative corrections due to Meister and Yennie ⁸⁾ using exponentiation of the doubly logarithmic terms. These formulae assume that the cross section is constant over the momentum acceptance of the detector system which is a reasonable approximation for the small angle data. The validity of this approximation was tested using formula (C.11) of Mo and Tsai ⁷⁾. Furthermore we compared the theoretical shape of the radiative tail, which was calculated using the correction formulae, with the measured spectra and found excellent agreement. The total corrections at all points amounted to between 12 and 15 %.

The cross sections are given in table 1. The errors assigned to the data points result from counting statistics after background subtraction, the uncertainty of the fluctuations in the target density ($\pm 0.5\%$) and the error resulting from the reproducibility of scattering angle and incident energy (± 0.5 mrad, ± 100 keV). To this we must also add the uncertainties in the normalization of the cross sections. These are due to the uncertainty in the solid angle of the second spectrometer ($\pm 1\%$), the absolute calibration of the current monitor ($\pm 1.5\%$) and the target thickness ($\pm 3\%$).

3. Data analysis

The measured cross sections have been analysed in two ways. First $G_E(q^2)$ and $G_M(q^2)$ were determined by Rosenbluth plots adjusting the Rosenbluth straight line by a least-squares fit to the experimental points at constant q^2 values. Some cross sections had to be slightly extrapolated to the proper value of q^2 . The results of the

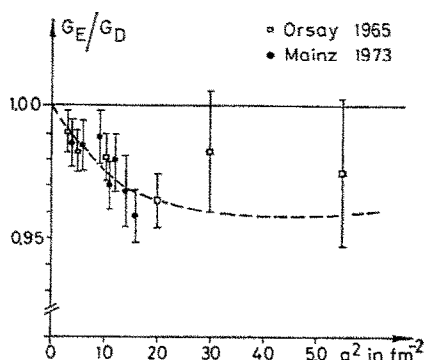


Fig. 1. The ratio of the electric form factor $G_E(q^2)$ to the predictions of the dipole fit $G_D(q^2)$ versus q^2 . The solid circles are the results of the analysis using Rosenbluth plots and the dashed line is the best fit form factor using eq. (4). The solid line at 1.00 on the ordinate of figs. 1–3 represents the dipole prediction.

analysis by Rosenbluth plots are shown as solid circles in fig. 1 for $G_E(q^2)$ and in fig. 2 for $G_M(q^2)$.

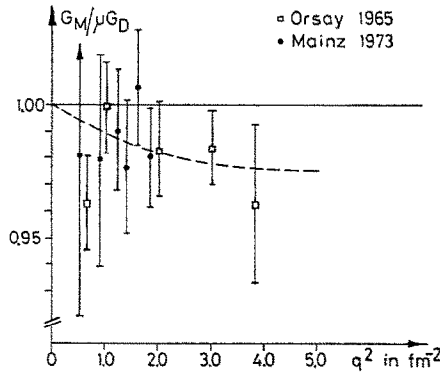


Fig. 2. The ratio of the magnetic form factor $G_M(q^2)$ to the predictions of the dipole fit $G_D(q^2)$ versus q^2 . The notation is the same as in fig. 1.

Secondly $G_E(q^2)$ and $G_M(q^2)$ have been determined by using analytical functions for the form factors in the Rosenbluth formula to fit the experimental cross sections $d\sigma/d\Omega_{\text{exp}}$. The dipole fit [eq. (2)] fails to describe our data. This is clearly seen from fig. 3 where the ratio $(d\sigma/d\Omega)_{\text{exp}}/(d\sigma/d\Omega)_D$ of the experimental cross sections to those predicted by the dipole fit and the form factor scaling relation is plotted against

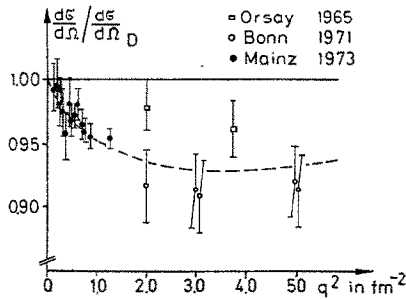


Fig. 3. The ratio $(d\sigma/d\Omega)_{\text{exp}}/(d\sigma/d\Omega)_D$ of the experimental cross sections to those predicted by the dipole fit and the form factor scaling relation versus q^2 for small scattering angles.

q^2 for small scattering angles. A reasonable description of our data is achieved by a fit which uses a sum of two double poles for each form factor and where no scaling of G_E and G_M is assumed, i.e.

$$G(q^2) = \sum_{i=1}^2 a_i / (1 + q^2/m_i^2)^2; \quad \sum a_i = 1. \quad (3)$$

As a final fit we used a sum of single poles for each form factor, i.e.

$$G(q^2) = \sum_{i=1}^4 a_i / (1 + q^2/m_i^2); \quad \sum a_i = 1, \quad (4)$$

because it is widely believed that the proton form factor should be dominated by a sum of terms $1/(1+q^2/m^2)$ where m is the mass of each of the known vector mesons. To get a satisfactory fit for eq. (4) four poles were necessary and again no form factor scaling was assumed. The parameters m_i of two of the poles were fixed at values corresponding to the masses of the ρ and ρ' mesons. The other parameters have been adjusted. To achieve a reasonable q^2 dependence of the form factor functions beyond our q^2 range our data have been analysed together with those measured at Bonn³⁾ and DESY⁴⁾. The following χ^2 function was minimized:

$$\chi^2 = \sum_{i=1}^m \left| \frac{\sigma_{\text{exp}}^i - \text{norm}_L \cdot \sigma_{\text{theor}}^i}{\Delta \sigma_{\text{exp}}^i} \right|^2. \quad (5)$$

The index i applies to the data points, $\Delta \sigma_{\text{exp}}^i$ is the random error of the cross sections. The parameter norm_L was introduced to check the normalization of the data sets of the different laboratories. The factors norm_L have been determined by assuming that $G_E(0) = 1$ and $G_M(0) = \mu$. The values of norm_L obtained in this manner are listed in table 2. They agree with the values found in recent fit calculations performed at DESY⁹⁾.

TABLE 2
The best fit values of norm_L

Lab.	Mainz	Bonn $q^2 \leq 20 \text{ fm}^{-2}$	Bonn $q^2 \geq 20 \text{ fm}^{-2}$	DESY
norm_L	1.003	0.981	1.008	1.003

4. Results and discussion

The best fit parameters for the form factors [eq. (4)] are listed in table 3. The best fit form factors are shown as dashed lines in figs. 1 and 2. The results of the analysis using Rosenbluth plots agree with the results using form factor functions.

TABLE 3
Best fit parameters for the form factors $G_E(q^2)$ and $G_M(q^2)$ as given by eq. (4)

	a_1	a_2	a_3	a_4	m_1^2 (fm^{-2})	m_2^2 (fm^{-2})	m_3^2 (fm^{-2})	m_4^2 (fm^{-2})
G_E	0.250	1.23	-0.442	-0.038	3.53	15.02*	44.08*	154.2
G_M/μ	0.843	0.522	-0.366	0.001	8.73	15.02*	44.08*	355.4

An asterisk denotes a fixed parameter.

The electric form factor $G_E(q^2)$ shows a significant deviation from the prediction given by the dipole fit (fig. 1). From the slope of the best fit electric form factor

[see eq. (1)] obtained from eqs. (3) and (4) we find a charge rms radius $\langle r_E^2 \rangle^{\frac{1}{2}} = 0.88 \pm 0.03$ fm and $\langle r_M^2 \rangle^{\frac{1}{2}} = 0.92 \pm 0.03$ fm, respectively. The errors of the radii result from the fitting routine according to eq. (5). The magnetic form factor (fig. 2) tends to be smaller than the prediction given by the dipole fit. The dipole fit is within the present experimental error, but our experiment at the forward angles is not very sensitive to the magnetic scattering. From the slope of the best fit magnetic form factor we find a magnetic rms radius $\langle r_M^2 \rangle^{\frac{1}{2}} = 0.83 \pm 0.07$ fm.

If we analyse with the same procedure the cross sections measured at Orsay ¹⁰⁾, Bonn ³⁾ and DESY ⁴⁾ disregarding the cross sections of this experiment we get results for the rms radii, mainly determined by the low q^2 data from Orsay, which agree within the limits of error with the values given above. Again the charge radius is larger than the radius for the magnetic moment distribution and the values obtained are 0.88 and 0.84 fm, respectively.

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